Coulomb + Newton: Towards a Cosmology

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Abstract

RealQM/RealNucleus models nuclei, atoms and molecules as *non-overlapping unit charge densities* — protons carrying the mass m_p and electrons the mass m_e — interacting by Coulomb potentials. Being continuum mechanics, the model has the same mathematical form as the models of the macroscopic materials those atoms build, so a hierarchy of continuum models over increasing scales follows from it. Adding **Newtonian gravitation**, which acts on the energy of that matter and matters only in the large, carries the hierarchy to galactic scales: one form of model from nuclei to the cosmic web. The reach is computational and not only formal — both ends are solved by the same primitive, a Poisson solve coupled to a relaxation of a real field on a Cartesian grid, so the gravitational sections are at the same time an application of the small-scale model's computational capacity, with the physics in them deliberately textbook. Cosmology is its natural arena, being the subject in which the nuclear scale meets the astronomical.

The chain is computed throughout. A small perturbation of the electric potential φ_E has its curvature read as charge, quantized by indivisibility into units of ± 1 in two spatial extents — the compact heavy proton, m_p, and the diffuse light electron, m_e. Coulomb builds nuclei, atoms and molecules from them. Newtonian gravitation then acts on the resulting energy density through its departure from the mean, $\nabla^2 \varphi_g = 4\pi G(\rho - \bar{\rho})$, a source of both signs although the matter is positive throughout, so the large scale separates into the overdense **cosmic web** and the underdense **voids**. The background source vanishes and the expansion **coasts**, $a \propto t$, $q = 0$. Only at the end is it recorded that the two forces are one two-valued Poisson form read with opposite signs; that parallel is used nowhere in the construction. Neither dark component is accounted for, and the large-scale inhomogeneity is an input.

Keywords: RealQM; RealNucleus; two-valued Poisson; Coulomb and Newton; charge quantization; density contrast; gravitation.

1 Introduction

RealQM formulates quantum mechanics as multiphase 3D continuum mechanics — unit-charge densities on non-overlapping domains separated by free boundaries, minimising the Coulomb energy [2], developed in the treatise *Real Quantum Mechanics* [1]. Its nuclear extension RealNucleus revives the pre-1932 proton–electron picture (neutron = proton + electron) and reproduces nuclear binding from pure Coulomb interaction, with no strong-force postulate [3]. Because these are continuum-mechanical models — densities on domains, forces from potentials, free boundaries where phases meet — they have the same mathematical form as the models of bulk matter, and a hierarchy over increasing scales follows with no change of mathematical character on the way up. Standard quantum mechanics, formulated in $3N$-dimensional configuration space, has no such continuum limit handing directly to the mechanics of materials.

This paper carries that hierarchy to its top. **Newtonian gravitation** acts on the energy of the matter RealQM builds and matters only in the large, so adding it extends the same model to galactic scales — a seamless extension, in that nothing about the mathematics changes: one more Poisson equation, one more force term in the same continuum mechanics. The hierarchy also extends *sideways* at the small scale, and the two directions should be distinguished. Coulomb is the static limit of **Maxwell**, and restoring the time dependence gives **Maxwell** + **Newton**, which brings **radiation** into the same framework — light as something the matter does (Section 7). This paper stays with Coulomb + Newton and notes where the wider pair would be needed. Where the extension leads is cosmology, and cosmology is its natural arena, being the subject in which the nuclear scale meets the astronomical. Stars shine by nuclear fusion: a process at 10^{-15} m sets the luminosity and chemical output of objects at 10^{9} m and thereby the evolution of galaxies at 10^{21} m. Light-element abundances are fixed by nuclear reaction rates yet read off the sky as cosmological data. And the dark sector, though *posed* at galactic scales, is *answered* at the smallest — dark matter sought as a new particle in particle colliders and underground detectors — so the instrument aimed at the largest missing mass in the universe is a machine that collides protons. A model that spans from nuclei to galaxies is therefore not an ambition imposed on the subject but what the subject already requires.

Why gravitation is here, and why it is deliberately textbook. A reader who reaches Section 9 will find nothing there that Newton did not have: the Poisson equation, a body force, a compressible gas. That is intended, for two reasons worth stating rather than leaving to look like a gap.

The first is that cosmology is not only the large scale. It is the whole chain, and its first question — where the protons and electrons came from — is a *small-scale* question, which is where this framework has something new to say: charge quantized by indivisibility, the neutron as $p+e$, nuclear binding from Coulomb geometry, two extents giving exactly two masses. Gravitation's job is to take that matter and gather it. If the gravitational sector were also novel, any success at large scales could be attributed to the new gravity rather than to the matter, and nothing would be learned. Keeping Newton textbook is what makes the test clean: **Cosmology** = **Coulomb** + **Newton**, with the novelty in what gravity acts *on*, and gravity itself the control rather than the claim.

The second is that the gravitational sections are a direct application of the *computational* capacity the small-scale model is built on, and are included as a demonstration of it. Nothing in that machinery is particular to Coulomb: it is a Poisson solve coupled to a relaxation of a real, non-negative field on a uniform Cartesian grid, with free boundaries located by the flow instead of imposed. Change the sign convention of the kernel and relax in time rather than downhill in energy and the same primitive is self-gravitating gas dynamics (Section 8). The claim being demonstrated is therefore not that gravitation needs revision but that one computational form, unchanged in its essentials, spans from a nucleus to a cosmic web — close to forty orders of magnitude in length — and does so on hardware as ordinary as a laptop or a browser. A method that could reach only the scale it was designed for would be a far weaker instrument.

The main idea, and the plan. *One small perturbation of one potential, followed upward by computation in one form of model, yields quantized charges and masses, then nuclei, atoms and molecules, and finally a web-like cosmos — and every step is computed, not assumed.* The continuity of that chain is the point, more than any one of its links, and the paper follows it in the order the model builds it. Section 2 states the exact model, both tiers. Sections 3–4 start the chain with the perturbation of φ_E; Sections 5–7 hand the resulting protons and electrons to RealQM and RealNucleus; Section 8 discretises both tiers; Section 9 adds gravitation on the energy density of the matter built; Sections 10–12 ask what more a cosmology would require — more than is delivered here. Only at the end (Section 13) is the parallel between the two forces recorded, as an observation about the finished construction. Security decreases along the chain and is marked as it does: the seed is a posit, the large-scale inhomogeneity an input, the cosmological claims weakest of all. Nothing along the way is static: the equilibria are balances maintained by motion, and the large-scale sector coarsens without ever settling.

2 The exact model

The macroscale model is **small-viscosity** (or **vanishing-viscosity**) compressible self-gravitating flow. With $\rho \geq 0$ the mass/energy density of matter, $\mathbf{m} = \rho\,\mathbf{u}$ the momentum, e the internal energy and p the pressure,

$$\begin{aligned} \dot\rho + \nabla\cdot\mathbf{m} &= 0,\\ \dot{\mathbf{m}} + \nabla\cdot(\mathbf{m}\,\mathbf{u}) + \nabla p + \rho\,\nabla\varphi_g &= 0,\\ \dot e + \nabla\cdot(e\,\mathbf{u}) + p\,\nabla\cdot\mathbf{u} &= 0, \end{aligned} \tag{1}$$

closed by $\mathbf{u} = \mathbf{m}/\rho$, $p = \gamma_s e$, $e = \rho\,T$, and coupled to gravitation through the Poisson equation with the **mean-subtracted** source,

$$\nabla^2\varphi_g \;=\; 4\pi G\,(\rho - \bar\rho), \qquad \bar\rho = \langle\rho\rangle. \tag{2}$$

Equation (1) is the Euler system with a gravitational body force. Its right-hand sides are zero because no dissipation is put in as a modelling choice; solutions are not smooth — the flow is turbulent and collapse forms shocks — so what is computed is a small-viscosity solution, the viscosity being that of the discretisation, mesh-dependent rather than physical. The meaningful output is therefore what is *stable* as that viscosity is decreased. This is a demanding criterion and we apply it to our own claims: it is what shows, in Section 9, that the arrest of collapse we appear to see is not yet established — the peak density keeps rising as the mesh is refined. Because ρ stays positive, $\mathbf{u} = \mathbf{m}/\rho$ is regular and no density floor or clamp is needed anywhere — unlike a genuinely signed density, for which $\mathbf{u}$ is singular at every zero crossing, precisely at the interfaces where the structure lives.

The $\bar\rho$ in (2) makes the gravitational source take both signs although the matter is positive throughout. It is not an added assumption: φ_g **exists only if its source has zero mean**, so some constant must be subtracted, and $\bar\rho = \langle\rho\rangle$ is the simplest such constant (Section 9).

The small-scale model is the Coulomb energy of RealQM: one non-negative density ψ_i per electron, each on its own domain Ω_i, the domains non-overlapping, $\Omega_i \cap \Omega_j = \emptyset$, separated by *free* boundaries. The state minimises

$$E[\{\psi_i\},\{\Omega_i\}] \;=\; \sum_i \int_{\Omega_i} \Big(\tfrac12|\nabla\psi_i|^2 - \sum_a \frac{Z_a}{|x-x_a|}\,\psi_i^2\Big)dx \;+\; \tfrac12\sum_{i\neq j}\iint \frac{\psi_i^2(x)\,\psi_j^2(y)}{|x-y|}\,dx\,dy, \tag{3}$$

subject to $\int_{\Omega_i}\psi_i^2 = 1$, the domains free to move: the interfaces settle where the energy says. The sum $i \neq j$ carries **no self-interaction** term — its role is taken by the localisation energy $\frac12\int|\nabla\psi_i|^2$ — and the problem lives in **three** dimensions however many electrons there are, one field each. Both (1)–(2) and (3) are discretised on the same kind of grid in Section 8.

3 The seed: charge and mass from a perturbation

In the standard account the origin of matter is an initial hot dense state, with inflation supplying the fluctuation spectrum. Here that role is played by a **small perturbation of the electric potential** φ_E, magnified by the Laplacian $\rho_E = -\varepsilon_0\nabla^2\varphi_E$ into $\pm$charge and hence into protons, electrons and all Coulombic matter. That is all the seed is asked to do. It creates the *matter*; it is not asked to lay out the large-scale *distribution* of that matter, and the gravitational sector makes no use of it.

The seed's amplitude at the creation scale is fixed rather than chosen. If φ_E oscillates there with wavelength λ and amplitude A, each half-wavelength is a domain of one sign carrying charge $q \sim \varepsilon_0 A\lambda$; demanding $q = \pm e$ exactly — one indivisible quantum per oscillation — gives

$$A \sim \frac{e}{\varepsilon_0 \lambda} . \tag{4}$$

At $\lambda \sim 1\,\mathrm{fm}$ this is $\sim 2 \times 10^7\,\mathrm{V}$, of the same order as the Coulomb potential $e/4\pi\varepsilon_0\lambda \approx 1.4 \times 10^6\,\mathrm{V}$ of the single charge it creates. The small-scale sector therefore reduces to **one unknown length** λ, the same unknown as the proton size (Section 14).

And with the charges, the masses. A cloud's energy is fixed by its extent (localisation cost against Coulomb attraction) and its inertia *is* that energy, E/c^2 (Section 7), so the two extents deliver **two inertial masses** and only two: the compact positive cloud is the heavy one, m_p, the diffuse negative one the light one, m_e. Quantization of charge carries quantization of mass with it — exactly two stable species because exactly two extents, so no intermediate mass appears any more than an intermediate charge does. Not settled here is the quantitative map: the observed $m_p/m_e \approx 1836$ is not reproduced by the naive inverse-size relation, which misses the proton's compactness by about a factor of ten. The *count* of masses is a result; their *values* are open (Section 14).

Two clarifications. The seed replaces the Big Bang as the origin of *matter*, not as a beginning of time: there is no initial singularity, and equally no account of why the seed is there or how it evolves — φ_E has no equation of motion here. And the large-scale distribution of matter is not derived from it: a statistically homogeneous seed leaves ρ uniform up to counting noise ($\delta \sim N^{-1/2} \sim 10^{-34}$, thirty orders of magnitude short of what structure requires), so the large-scale inhomogeneity is an **input** to the gravitational sector, not an output of the electric one. Whether one seed could supply both — which would need its long wavelengths to modulate its short-wavelength power, i.e. local-type non-Gaussianity — is left to Section 14.

4 The arena

The equations are written in a **3D Euclidean coordinate system** — one of many possible, and nothing depends on which. It is a system of coordinates in which to express the fields, not an absolute space and not a physical medium; no ether is postulated and no frame is privileged. The only operator needed is the Laplacian. Charge is not postulated separately but is the curvature of the potential, $\rho_{\text{charge}} = -\varepsilon_0 \nabla^2 \varphi_E$, with two immediate consequences. **Both signs:** ρ_{charge} is positive where φ_E is concave, negative where convex. **Net zero:** $\int \nabla^2 \varphi_E \, dV$ is a surface term that vanishes, so the universe carries zero net charge automatically — equal numbers of protons and electrons, with no baryogenesis, and once the neutron is read as a proton–electron pair, in the far stronger *local* form of Section 5. Because the Laplacian multiplies each Fourier mode by k^2, a short-wavelength oscillation becomes a large charge density.

5 The small scale: RealQM and RealNucleus

The seed has delivered charge in both signs, quantized at ± 1, in two spatial extents: small dense positive and large diffuse negative charge, the **proton** and the **electron**. From here the whole charged world is built by (3), by Coulomb geometry alone.

Energy, not mass. In RealQM a particle is a charge cloud whose size is fixed by localisation cost against Coulomb attraction, and its inertia is its energy content E/c^2 — a *measured* identity (the Penning trap, the nuclear mass defect), not a relativistic postulate [4]. No primitive "mass" is invoked: charged matter carries energy, and energy gravitates. A contracted cloud costs more, so small clouds are energy-dense — the small proton is the heavy one. Whether size fixes energy *quantitatively* is another matter, left open; the cosmology needs only that gravity couples to energy.

Self-repulsion is kinetic energy. RealQM clouds do not overlap and carry no self-interaction. The role self-repulsion would play, resisting collapse to a point, is carried by the localisation energy $\frac{1}{2}\int |\nabla\psi|^2$: compressing a cloud sharpens its gradients and costs kinetic energy, so it settles at finite size. A free electron spreads; a bound one takes the size where localisation balances attraction.

Charge is quantized: ± 1 the only possibility. To exclude an electron's Coulomb energy with *itself*, the "self" must be well defined: there must be an *indivisible* entity whose interior is left out of the Coulomb sum. A divisible continuum would make "self" ambiguous — is half a cloud a self? do the halves repel? — and the prescription ill-posed. So removing self-repulsion *requires* indivisible units. A unit is identified by its spatial occupation, and for all electrons, and equally all protons, to be the same entity, each domain must carry the same charge. Every electron then carries -1 and every proton $+1$, and no intermediate charge can exist. Granted that the no-self-interaction law is one *universal* law rather than an ad-hoc per-blob subtraction, quantization is a consequence, not a postulate — where the Standard Model must *import* it, anomaly cancellation fixing only the $\pm$ ratio and true quantization needing a Dirac monopole or a compact unifying group, both unobserved. What this does not fix is the *magnitude* of the unit, nor the fractional charges of confined quarks, outside RealQM's no-substructure scope.

Nuclei and atoms, one mechanism at two scales. With a neutron taken as $p + e$, a neutral atom (A, Z) holds A protons and A electrons: $A - Z$ *compact* inside the nucleus and Z *expanded* in orbitals. The electron occupies two size-phases, its extent adapting to the well: compact in a deep one (nuclear, ∼fm, ∼MeV), expanded in a shallow one (atomic, ∼Å, ∼eV). Chemistry and nuclear physics are then one Coulomb free-boundary mechanism at two scales: a molecule is nuclei glued by expanded electrons, a nucleus is protons glued by compact ones. RealNucleus reproduces nuclear binding this way — the He-4/deuteron binding ratio comes out 13.1 against an experimental 12.7, from Coulomb geometry alone [3] — and light-nucleus fusion appears as a rearrangement of protons and electrons, not the action of a separate force.

Neutrality is local. Because a neutron *is* a bound proton–electron pair, every neutron carries its own compensating charge: matter is neutral *pointwise*, not merely on average. Here that is automatic twice over — globally from the net-zero seed, since $\int \nabla^2 \varphi_E \, dV$ is a surface term that vanishes, and locally because a neutron cannot unbalance a count of $\pm$ pairs.

What this sidesteps, and what it does not. It is worth being exact about the comparison, because the corresponding problem in the standard account is a hard one. There, matter originates in a hot plasma of particles and antiparticles, and its survival requires **baryogenesis**: Sakharov's conditions — baryon-number violation, C and CP violation, and departure from thermal equilibrium — must all be met in the early universe. The Standard Model contains all three in principle and fails on two quantitatively. The CP violation available in the CKM matrix is smaller than required by something like ten orders of magnitude, and for the observed Higgs mass the electroweak transition is a *crossover* rather than first-order, so the departure from equilibrium is not there either. The observed baryon-to-photon ratio $\eta \approx 6 \times 10^{-10}$ is consequently *not* accounted for by the Standard Model, and supplying it is among the standard motivations for physics beyond it.

In the present picture no such epoch is posited: charge appears in $\pm$ pairs as the curvature of one potential, so the balance is a property of the arithmetic rather than a residue left by a near-perfect cancellation, and nothing has to be fine-tuned to one part in 10^9 to leave the matter we see. But the sidestep should be named for what it is. The standard problem concerns *matter versus antimatter*, whereas the two signs here are the *proton and the electron*, both matter — so the asymmetry question is **dissolved rather than answered**, and that comes at a price. Antimatter exists: positrons are produced in the laboratory and observed astrophysically. A positive charge cloud of *electron* extent is exactly what a positron would be in this framework, so nothing forbids one, and the model owes an account of why the universe is not populated with them — the counterpart, in this language, of the question it has avoided. We do not have one, and it belongs with the open problems of Section 14.

The comparison is also computational. Where the standard account is quantitative about bound matter it is so through machinery of a quite different weight. On the nuclear side the predictive instrument is lattice QCD: a Monte Carlo evaluation of a path integral over quark and gluon fields on a discretized four-dimensional spacetime, run as supercomputer campaigns of millions of core-hours with community codebases of order 10^5 lines. On the atomic and molecular side the many-electron Schrödinger equation lives in $3N$ dimensions and is never solved as written; what is solved is a hierarchy of approximations to it — Hartree–Fock, coupled cluster at $\mathcal{O}(N^7)$, density functionals whose exchange–correlation part is in practice fitted — inside large packages that few of their users read. The model here is at the other extreme: one real, non-negative density per electron in ordinary three-dimensional space, a Poisson solve coupled to a relaxation on a Cartesian grid, cost linear in electron number, no basis set and no fitted parameter. The atom solver used for this work is 350 lines of JavaScript and runs in a browser tab; the script behind Figure 1 is 115 lines of Python. The whole chain of this paper is computed by code a reader can read. That this holds at *both* ends is not economy of effort but a structural fact: Coulomb and Newton are the same computational problem, a discrete Laplacian inverted against a two-valued source, so the same solver serves a nucleus and a cosmic web and no separate large-scale machinery has to be built (Section 8).

Simplicity is not accuracy, and the claim is only the former. Lattice QCD delivers the hadron spectrum from first principles at the percent level, which the present model cannot do at all — the absolute MeV scale is its open problem (Section 14) — and coupled cluster reaches a chemical accuracy not matched here. Those codes are complicated because the formulations behind them are, and they buy real results with it. What a short and inspectable code buys instead is of a different kind: every number in it can be traced to the equation that produced it, there is no fitted function absorbing the discrepancy, and any result can be rerun and broken by a reader in an afternoon. That is a virtue for a model asking to be tested, not evidence that it is right.

6 Complexity, and the two domains

The world is built not on a charge imbalance (charge is exactly balanced) but on a *size* imbalance: the proton has a fixed small size while the electron's is variable, adapting to its well. The fixed proton supplies anchors; the variable electron *spans scales* — spreading to bind atoms, compacting to bind nuclei — producing the nested hierarchy that *is* complexity. With a single fixed size only one scale exists and nothing complex is built, so the nuclear/atomic separation is a **precondition for complexity**: an anthropic reason why the scales are separated, though not a derivation of how far. Since a neutron is structurally $p + e$ it is heavier than a proton by construction, so proton and hydrogen stability is automatic — the usual fine-tuning bound "α must not be too large or the proton decays" does not arise.

Why the proton must be compact. A caged electron, held by several protons, sits at zero localisation and takes the proton size R_p; a free one takes the atomic size a_0, whence $a_0/R_p \approx 1/\alpha^2$: the two domains barely overlap because the two branches are separated by $\alpha^{-2} \approx 2 \times 10^4$. The picture also states its own validity limit. Since inertia is energy, a compact electron bound by Coulomb at the nuclear scale has binding $\sim e^2/R_p$, and requiring it to stay below the electron's own energy E_e gives $R_p \gtrsim e^2/E_e$: a floor under the proton size, a plain comparison of two measured energies needing no appeal to α. Nature sits essentially *at* the floor — at the deuteron radius ($\sim$2.1 fm) the Coulomb binding $e^2/R_p \approx 0.7$ MeV already exceeds $E_e \approx 0.5$ MeV — so the nuclear scale saturates the theory's own validity limit, which is why β-decay is where the static-cloud description is most stretched.

7 Inertia is energy; the local speed limit

Throughout, inertia is energy content: $\mathbf{p} = (E/c^2)\mathbf{v}$, the measured relation of the Penning trap [4], with $\mathbf{F} = \dot{\mathbf{p}}$ governing the response. In the small the force is electromagnetic and couples to *charge*; in the large it is gravity and couples to *mass* $= E/c^2$. One mechanics, one inertia, two forces reading two properties of the same matter.

The constant c enters *only* the electromagnetic sector, through the field momentum of a moving charge. Hence a **local speed limit**: feeding energy to a charge only increases its inertia E/c^2 without bound as $v \to c$. But Newtonian gravitation is c-free and imposes no speed limit on the large-scale motion of neutral bodies: two galaxies may recede at relative speed $> c$, a global relative velocity of gravitating matter rather than a local signal, violating no local causality — as in observational cosmology, where distant galaxies recede faster than light without any local law being broken.

Coulomb is the static limit; Maxwell adds radiation. What is used throughout this paper is *electrostatics* — Coulomb's law, the t-independent first member of the electromagnetic pair. That is a restriction of convenience, not of principle, and the rung above is worth naming because it is the one that adds the missing physical process. Replacing Coulomb by the full **Maxwell** equations extends the same continuum framework to time-dependent charge and current, and with them to **radiation**: the model becomes **Maxwell + Newton**, and light ceases to be an ingredient supplied from outside and becomes something the matter *does*. Nothing in the mathematics changes character here either — fields on the same space, sources from the same charge densities — so this is a continuation of the hierarchy rather than a departure from it.

The gain is not decorative. Two of the three epochs of Section 11 are *defined* by radiation: nuclei form and the Coulomb binding energy leaves as light; atoms form and the matter turns transparent, releasing the radiation that had been trapped. In the present electrostatic treatment those steps are described by their energy bookkeeping alone — so much binding energy released — with no account of the emission itself. Maxwell supplies the missing half, and it is also what any eventual treatment of the microwave background would have to be built on. The time-dependent form of RealQM required for this, in which a real charge density carries a *complex* current and a radiating dipole, is developed in [5]; nothing in the present paper depends on it, and we make no claim here about radiation beyond noting where it enters.

8 The computational model

Both tiers are solved by the same numerical primitive: a Poisson solve coupled to a relaxation on a uniform Cartesian grid, ∇^2 the standard 7-point stencil, the matter field then advanced — downhill in energy at the small scale, in time at the large one. No basis sets, no wavefunctions in $3N$ dimensions, no fitted parameters.

	Small scale (Coulomb)	**Large scale (Newton)**
unknown	densities $\psi_i \geq 0$ on domains Ω_i	contrast ρ, momentum m,
field solve	$-\nabla^2\varphi = \rho_{\text{charge}}/\varepsilon_0$	$\nabla^2\varphi_g = 4\pi G(\rho - \bar{\rho})$
solver	multigrid V-cycles, Jacobi smoother	red–black SOR, $\omega = 2/(1 +$
relaxation	steepest descent on the Coulomb energy	explicit compressible flow i
free surfaces	interfaces $\partial\Omega_i$ move with the energy	the $\rho = 0$ contact wall
typical grid	one global lattice, 200^3 for a large molecule; $\sim 10^3$ cells per electron domain	$200^3 = 8 \times 10^6$ cells

Why one code covers both, and exactly how far the sameness goes. The two columns are not similar by arrangement. Read the *field solve* row: the two equations are one equation, differing in the sign of the source and in a constant. Coulomb and Newton pose the *same computational problem* — invert a discrete Laplacian on a grid against a two-valued source — so the expensive and nontrivial half of the work is literally one routine, and it neither knows nor cares whether the grid spacing is a femtometre or a megaparsec. That is why there is no separate large-scale method to develop, and why the code stays small while the physics changes scale by forty orders of magnitude. It is also, stated computationally, the observation recorded formally in Section 13; nothing in the construction takes it as a premise, but the solver has been using it throughout.

The sameness stops in a definite place, and it is worth marking. What differs between the columns is the *relaxation*: steepest descent to a static minimiser at the small scale, explicit time-stepping of a compressible flow at the large one. That difference is one of question — equilibrium versus evolution — not of scale or of force, and a static gravitational configuration or a time-dependent Coulomb problem would swap the two without touching the field solve. The choice of iteration (multigrid against red–black SOR) is likewise implementation and history: either solves either, and the pairing above is what each tier was written with, not what it requires.

At the small scale the cost grows *linearly* in electron number where a wavefunction method grows exponentially, and the Neumann-free interface — located by the flow rather than imposed — lets one code describe an atom, a molecule and, with the electron compacted, a nucleus. These solvers run in the browser (WebGPU).

At the large scale, since the model is small-viscosity flow the scheme supplies no physical viscosity: the computation *is* the regularisation, and a result counts only if it persists as that regularisation is weakened. Two numerical points carry physics. First, the velocity is taken from $|\rho|$, $u = m/(|\rho| + \rho_{\min})$, so **inertia is positive in both phases** while the sign enters only the gravitational coupling — the direct expression of the contrast reading, underdense regions being ordinary matter of which there is less than average. Taking the velocity from signed ρ instead produces the classic runaway: the negative phase cannot self-support and is consumed, the void fraction collapsing from 50% to a few percent. Second, transport must be conservative and upwind; with centred differencing the web survives but is over-diffusive and the filaments never sharpen.

What the computations establish. At the small scale they are quantitative: binding energies, ionisation energies and nuclear binding follow from the same free-boundary minimisation without adjustable parameters. At the large scale they are *morphological only* — dimensionless units, a periodic box, no expansion, no redshift, no calibration to any survey — and nothing in them tests the cosmological claims of Sections 10–12. As *physics* that is a weak claim, and it is not strengthened here. As a demonstration of the *method* it is a strong one, and the two should not be confused: an 8×10^6-cell self-gravitating flow and a free-boundary atom are the same solver on the same grid at scales some forty orders of magnitude apart, with no basis sets, no fitted parameters and no change of formalism between them.

9 The large scale: Newtonian gravitation

Nothing in this section is new, by design (see the Introduction): it is Newtonian gravitation as it has always been, acting on the matter the preceding sections built. What is being carried over from the small scale is not physics but *method* — the same grid, the same Poisson solve, the same relaxation of a real field. The mass/energy density of that matter — protons, electrons, their binding and radiation — is ρ, positive throughout. It is the source of the second force, but enters its Poisson equation only through its departure from the mean, (2), so the *source* takes both signs — positive where matter is denser than average, negative where sparser — while the matter itself is positive everywhere. There is no negative mass and no second species: the two signs are *excess* and *deficit* of one substance.

Why the subtraction is forced, and what is chosen. The zero mean is not an assumption added to Newton's law; it is what makes φ_g exist. On an unbounded (or periodic) domain the Poisson equation $\nabla^2\varphi_g = 4\pi G\sigma$ is solvable only if $\langle\sigma\rangle = 0$: for a uniform positive source the potential diverges, and the net force on a point depends on the order in which the infinite distribution is summed — the Neumann–Seeliger difficulty, which is not a peculiarity of this model but of Newtonian cosmology as such. Some constant must therefore be subtracted, and no Newtonian cosmology escapes it. Standard practice supplies it as bookkeeping and calls it the Jeans swindle.

What is *chosen* is only which constant, and $\bar\rho = \langle\rho\rangle = M_{\text{tot}}/V_{\text{tot}}$ is the simplest — the one that makes the total gravitational charge of the universe vanish, matching the zero net charge of the Coulomb sector. Two consequences follow. First, $\bar\rho$ is a boundary condition on the universe as a whole, so the local force law refers to a globally fixed constant: the model is non-local in this one limited sense, and a mean over some finite radius R would instead make the dynamics depend on an arbitrary R. Second, $\bar\rho$ is time-dependent — under expansion it dilutes as a^{-3} with the matter, so the cancellation is preserved at every epoch.

The non-standard step is thus not the subtraction but the refusal to treat it as bookkeeping: here the mean genuinely carries no gravitational charge, and the consequences are followed to the end (Section 10). This is where the model parts company with general relativity rather than with Newton, since in general relativity a homogeneous background *does* curve spacetime and the existence argument above does not apply — the solvability difficulty is Newtonian. What that costs is stated in Section 12.

This is *not* modified gravity but a modified source. The force law is untouched — no altered $1/r^2$, no extra field, no acceleration-dependent coupling — and for a galaxy, where the cosmic mean lies thirty orders of magnitude below the local density, subtracting $\bar\rho$ changes nothing. What is non-standard is what counts as a source at the largest scale. The accurate description is *standard Newtonian dynamics with a non-standard cosmological boundary condition.*

The force-rule. Under **like sources attract, unlike repel**, all four combinations are realised and each is observed. Two overdense regions attract: clustering. Two underdense regions attract: *voids merge*, as they are seen to. An overdense and an underdense region repel: matter drains from voids into filaments. The large scale therefore separates into overdense and underdense domains — the luminous **cosmic web** of filaments and clusters, and the **voids**, which are not inert leftovers but push their surroundings outward. The asymmetry with the charge sector is worth marking: a positive charge can be isolated, whereas the two signs here are excess and deficit of one substance, meaningless without the mean to measure against. That is why charge comes in indivisible ±1 units carrying an identity while gravitational "charge" is continuous and reference-dependent; the quantization argument belongs to the Coulomb sector and must not be carried over.

What sets the scale, and what arrests collapse. The web scale is the **Jeans length** $\lambda_J = 2\pi c_s/\sqrt{G\rho}$ with $c_s = \sqrt{\gamma_s T}$: a box below one Jeans length is pressure-supported and inert. What *arrests* collapse is the adiabatic index $\gamma_{\rm ad} = 1 + \gamma_s$ through $M_J \propto c_s^3/\sqrt{\rho}$: under adiabatic compression $c_s^2 \propto \rho^{\gamma_{\rm ad}-1}$, so $M_J \propto \rho^{3(\gamma_{\rm ad}-1)/2-1/2}$, rising with density when $\gamma_{\rm ad} > 4/3$ and falling below it. Above 4/3 collapse shuts itself off at finite scale; below, each collapsing region keeps fragmenting. (An earlier version attributed the arrest to an artificial shock viscosity. Numerical experiment does not support that: the shock-heating term destroys the web in a thermal runaway and worsens rather than converges under timestep refinement.)

Web formation is standard, and illustrated rather than claimed. Equations (1)–(2) are ordinary Newtonian self-gravitating gas dynamics with a mean-subtracted source; nothing in them is new. That such systems develop a filamentary web from a Gaussian seed is classical: by the Zel'dovich approximation [8] collapse proceeds anisotropically, since the tidal-tensor eigenvalues generically differ, so a region contracts first to a sheet, then a filament, then a node — the morphology made quantitative by Hessian-eigenvalue classification [9]. Figure 1 shows the model's own equations doing this, in a coasting background and with $\gamma_{\rm ad} = 1.4 > 4/3$, so that the arrest criterion above is satisfied.

Three things in that run are worth stating, since each was checked rather than assumed. *The arrest is **not** demonstrated, and we say so.* At any one resolution peak/mean does stay bounded near 10^3 at $\gamma_{\rm ad} = 1.4$, against 1.3×10^4 and climbing for the same seed at $\gamma_{\rm ad} = 1.2 < 4/3$ — which looks like the Jeans-mass criterion doing its work. But a resolution ladder on an identical seed at fixed time ($N = 64, 96, 128, 160$, the seed Fourier-restricted to each grid so the physical field is the same) gives peak/mean 650, 912, 1232, 1616 — a factor 2.49 for a factor 2.50 in resolution, i.e. $\rho_{\rm peak}/\bar{\rho} \propto N^{1.00}$. The bound is therefore the *mesh*, not the physics. The exponent also says what is happening: $\rho_{\rm peak} \propto 1/h$ is the signature of a *sheet* one cell thick — a Zel'dovich pancake whose caustic the grid cannot resolve — and not of a collapsed core, which would steepen far more. Consistently, the Jeans length at the density peak stays at 2–3 cells at every resolution, short of the $\sim$4 needed to resolve a collapse, and refining does not improve it because the peak density rises to cancel the gain ($\lambda_J \propto \rho^{-0.3}$).

The volume statistics behave better and are quoted on that basis: over the same ladder the deep-void fraction runs 41.7, 46.5, 49.3, 51.3 per cent, with increments decreasing roughly geometrically towards $\sim$55 per cent, and the concentration measure likewise. Morphology is thus good to a few per cent while peak densities are not converged at all. The analytic argument for arrest is untouched by this — $M_J \propto \rho^{0.1}$ does rise for $\gamma_{\rm ad} > 4/3$ — but the exponent is so small that demonstrating it numerically needs resolution we do not have. What the runs do show is the *difference* between the two indices, which is robust across the ladder. *The web is nevertheless a transient*: it forms, sharpens, and then *coarsens*, small nodes merging into large ones until little web is left, so the morphology has a window rather than a limit. And *a non-expanding box is worse, but not for the expected reason.* It builds a web and then destroys it thermally: with no expansion to drain the heat of compression, pressure unwinds the structure and the deep-void fraction falls from 87% back to $\sim$0% with peak/mean returning to $\sim$10. Expansion is thus not needed to keep the box from collapsing — a zero-mean source exerts no net force on a uniform state, so a static box is self-consistent here in a way it is not in ordinary Newtonian cosmology — but it *is* needed to keep a web from blowing itself apart. The two roles should not be confused.

A caution about reading such runs, and it cuts both ways. Volume statistics and visual impression disagree, and neither is sufficient alone. In the run of Figure 1 the deep-void fraction and the mass concentration both climb monotonically — to 73% and 80% — while the picture is becoming *less* web-like, because concentration keeps rising as the web coarsens into isolated clumps. Conversely a thin bright filamentary image can occupy very little volume and be classified almost entirely void. A further trap is instrumental: a Hessian-eigenvalue classifier thresholded on the *r.m.s.* curvature reports the field as structureless exactly when it is most clumped, since a handful of extreme peaks then set the threshold and genuine ridges fall below it (a median-based threshold does not do this). Web-like appearance, web-like statistics and web-like classification are three different claims.

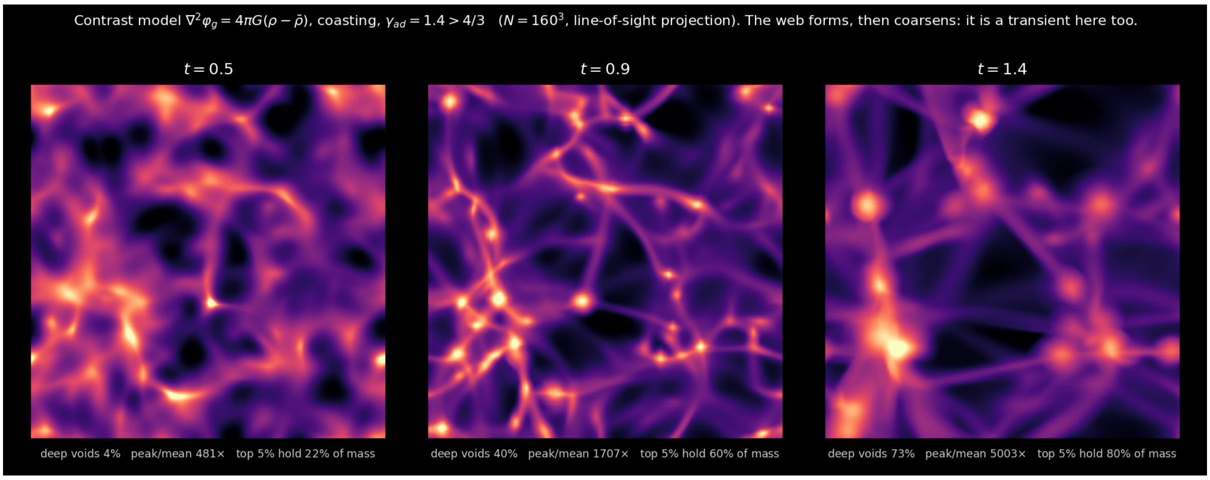

Figure 1: Gravitational instability of the contrast source (2) from a Gaussian seed, coasting background, $\gamma_{\rm ad} = 1.4 > 4/3$ (160^3, line-of-sight projection, logarithmic scale; bright = overdense). Filaments meeting at nodes with deep voids between, then coarsening. Total mass is conserved to 10^{-5}. The volume statistics drift by a few percent per refinement and the peak density not at all convergently (see text), so the morphology is indicative and the peak values are upper bounds set by the mesh. The morphology is the classical Zel'dovich sequence [8] and is *not* particular to this model; it illustrates the dynamics and is not offered as evidence for the cosmology.

What the seed of Figure 1 is, and why it is not innocent. The seed there is a Gaussian random field with a *single correlation length*, placed above the Jeans scale. That choice is doing more work than a figure caption usually admits, and the alternative was tried. Seeding instead with the scale-free $P_\rho \propto k$ of Section 10 produces *no structure at all*: at equal total variance the run reaches peak/mean 1.8 against 912, with no deep voids, because $P \propto k$ carries power per logarithmic interval $\propto k^4$, so nearly all the variance sits *below* λ_J where pressure forbids growth. Renormalising instead to give the same power *at* λ_J does not rescue it: the required total variance is $\sim$26 times larger, which drives the density negative over 46% of the volume, and the seed is then not a perturbation but a saturated field.

That leaves the question the two runs were circling, which is not whether the seed collapses but whether it carries the observed *shape*. The matter power spectrum rises as $P(k) \propto k^{n_s}$ with $n_s \approx 0.96$ on the largest scales, turns over, and falls beyond. Measuring $P(k)$ directly from these runs (spherically averaged in k-shells, at $N = 96, 128, 160$, all at $t = 1$) gives a consistent and unfavourable answer:

seed	n on $k = 2$–8, seed $\to$ evolved	k of the peak of $k^3 P$	peak/mean
Gaussian bump (Figure 1)	$-1.5 \to -1.0$	16–18, stable in N	870–2900
scale-free, $P_\rho \propto k$	$+0.9 \to -0.9$	24–49, wandering	1.7–2.0
observed	$\approx +1$	—	—

Neither seed delivers it. The one that collapses has the large-scale slope wrong by about two in the exponent, with too much power on large scales rather than too little, and its web scale is simply its own correlation length — $k \approx 17$, fixed by the seed and unchanged as the grid is refined, so the characteristic scale of the pattern is put in by hand rather than selected by gravity. The one with the correct large-scale slope does not collapse at all. A control run with the gravitational coupling set to zero shows in addition that the late-time slope is not the seed slope in either case: pressure smoothing decays the small scales and leaves relatively more power at large ones, an effect gravity partly opposes rather than causes. So a seed cannot simply be handed the observed spectrum and be expected to deliver it downstream. The script is `power_spectrum.py` in the numerics directory.

A raw scale-free spectrum is therefore unusable as an initial condition here, in either direction. That is not a peculiarity of this model — it is why standard cosmology processes the primordial spectrum through a **transfer function**, which suppresses the small-scale power before structure formation begins — but this model has no transfer function (Section 10), and the single-scale bump of Figure 1 is in effect standing in for one. The figure should therefore be read as showing that *these equations make a web from a suitable seed*, not that this model produces the observed web from initial conditions it can justify.

Against the observed web the run gets the voids about right and the rest wrong: at the last panel 73% of the volume is deep void (observed $\sim$60–80%), but filaments and walls hold only $\sim$5% and $\sim$3% of the volume against an observed $\sim$10–20% and $\sim$20–30%. The matter sits in structures too thin.

The honest debt runs the other way. Standard simulations reproduce the observed web using *cold dark matter* in an *expanding* background, neither available here. Baryons alone are the harder case: before recombination they are coupled to the radiation and cannot grow, and gas pressure suppresses structure below the Jeans scale — which is precisely the structure-formation argument *for* dark matter, and among the strongest. That these equations admit webs is no answer to it.

Mathematically the mechanism is a **phase separation** driven by the long-range gravitational kernel rather than by local diffusion — a nonlocal Cahn–Hilliard [15, 16] whose kernel is attractive for like sources, so that coarsening is driven rather than frustrated. One consequence matters observationally: the double well is *not* symmetric, since $\delta \geq -1$ while overdensities grow without bound, so filaments and voids are not statistical mirrors. That symmetry *would* hold for a genuinely signed mass density, and the asymmetry of real voids counts against that reading.

10 Voids, the background, and the spectrum

Void repulsion is real but is not dark energy. A region below the mean is a negative source in (2), so it pushes matter outward and grows: voids are actively driven, not merely evacuated. But this is a local, peculiar effect, present in any mean-subtracted gravity, and must not be confused with dark energy, whose defining role is *accelerated expansion of the background.* There the model is explicit and negative: with the mean not gravitating the Friedmann source vanishes, $\ddot{a} = 0$, and the expansion coasts ($q = 0$) rather than accelerating ($q_0 \approx -0.55$). **The model does not reproduce what dark energy was introduced to explain.** What it offers instead is that no Λ need be introduced and that the coincidence problem does not arise: the cancellation is exact at every epoch rather than mysteriously near-exact now. Acceleration is *excluded*, not merely absent — both sides of the contrast are the same substance and dilute together as a^{-3}, so the cancellation holds for all time; acceleration requires $w < -1/3$ while exact cancellation forces $w = 0$.

Negative mass as a substance: set aside. An earlier form of this model, and work by others [7, 10, 11], takes the negative sector to be a substance rather than a deficit. We do not follow that route: such a mass would be a species never detected, it would *lens* with the opposite sign whereas void lensing is measured and consistent with ordinary underdensity, and a signed density makes $\mathbf{m}/\rho$ singular at every zero crossing, forcing regularizations that dominated the result in our own experiments. The deficit reading needs none of them.

Inflation, and the spectrum. The Laplacian amplifies short wavelengths, $\rho_k \propto k^2 \varphi_k$: an inflation of the density *field*, not the exponential expansion of *space*, which addresses the horizon and flatness problems and is not reproduced here. Within the electric sector this has a tidy consequence — one might expect the k^2 weighting to leave nothing at large scales, and it does not, since in power $P_{\rho_E}(k) = k^4 P_\varphi(k)$, so a potential with equal fluctuation power per octave ($P_\varphi \propto k^{-3}$, requiring no tuning) gives $P_{\rho_E} \propto k$.

It is tempting to read that as delivering the primordial *matter* spectrum, and earlier versions of this paper did. It does not, and the reason is the same one that runs through Section 3: $\rho_E = -\varepsilon_0 \nabla^2 \varphi_E$ is the **charge** density, and charge screens to neutrality precisely at the scales where the matter spectrum lives. Carrying the result across would require the seed's long wavelengths to modulate its short-wavelength power — the very link this paper does not assume (Section 14). **The large-scale matter spectrum is therefore an input here, not a result**, its shape included; the simulations of Section 9 impose it. What the calculation above establishes is only that the electric sector carries power at every scale rather than collapsing onto the finest, which is what the k^2 weighting might have suggested.

11 The cosmic sequence and the helium fraction

The two sectors exchange energy in both directions: the energy of the charged world is mass and sources gravity, while collapse converts gravitational potential energy into the heat that drives the electromagnetic processes. As contracting matter heats, nuclei form (∼MeV) and the Coulomb binding energy radiates away as the mass defect; on cooling, nuclei capture expanded electrons into atoms (∼eV), the matter turns neutral, electromagnetism screens out and radiation streams free. Only then does gravitational aggregation build the web. A bound state survives once the bath can no longer ionise it, so the two electron phases switch on at temperatures set by their binding energies, forcing the order plasma → nuclei → atoms.

epoch	process	temperature
plasma	free ± charges	$> 10^{10}$ K
nuclei	compact e captured → nuclei	$\sim 10^{10}$ K ($kT\sim$MeV)
atoms	expanded e captured → atoms; radiation	~ 3000 K ($kT\sim$eV)

Table 1: Two condensation epochs, separated by $\sim 10^6$ in temperature.

The helium fraction: a consistency, not a prediction. The electron's two size-phases — compact (inside a neutron, since neutron = p + compact e) and expanded (atomic) — are interconverted by $p + e \to n$ and $n \to p + e$. Three quantities enter and all three are *inputs*: the gap $E_* \approx 1.29$ MeV between the phases, identified with the neutron–proton mass difference (its absolute size is the open nuclear-scale problem, RealNucleus fixing ratios and not the MeV scale); the transition temperature $T_* \sim E_*/k \sim 10^{10}$ K; and the freeze-out temperature $T_f \approx 0.8$ MeV, set as in standard big-bang nucleosynthesis. In equilibrium n/p is the Boltzmann factor $e^{-E_*/kT}$, locking at freeze-out; a fraction β-decays back (survival factor d) and essentially all the rest end in He-4:

$$\left(\frac{n}{p}\right)_{\rm nuc} = \frac{(n/p)_f\, d}{1 + (n/p)_f(1-d)}, \qquad Y = \frac{2\,(n/p)_{\rm nuc}}{1 + (n/p)_{\rm nuc}}. \tag{5}$$

With $E_* \approx 1.29$ MeV, $T_f \approx 0.8$ MeV and $d \approx 0.74$: $(n/p)_f = 0.199 \to 0.140 \to Y = 0.245$, the observed primordial helium. Nothing here is derived; RealNucleus's one genuine computed output, the He-4/deuteron binding ratio, does not enter Y. The content is that the proton–electron reinterpretation is *consistent* with the standard calculation.

12 Towards a cosmology: what would be required

Shared with the standard account: ordinary matter is Coulombic and gravitation attractive between masses; nuclei and atoms form and later assemble into filamentary structure; $Y \approx 0.25$ is accommodated. The comparison below is deliberately narrow. Gravitation is textbook here by design, so it is not a point of difference, and where the standard account reaches past Newton — relativistic cosmology, the dark sector, the thermal history — it is reaching outside what this paper claims, which cannot be scored as a disagreement. Two axes remain, and they are the two where the accounts genuinely differ: *where matter comes from*, and *what it takes to compute it.*

	Standard account	**Here (RealQM/RealNucleus)**
1. The origin of matter		
origin	hot plasma of particles and antiparticles; survival requires a baryogenesis epoch	$\pm$ pairs as the curvature of one potential; no such epoch posited
matter/ antimatter	Sakharov's conditions; CKM CP short by $\sim$10 orders, the electroweak transition a crossover; $\eta \approx 6 \times 10^{-10}$ unexplained	question *dissolved, not answered*: the two signs are proton and electron, both matter. Positrons are then owed an account, and none is given
charge quantization	assigned; constrained by anomaly cancellation, not derived	a consequence of the indivisibility of a unit cloud: ± 1 the only possibility
the neutron	elementary bound state of quarks (udd)	composite: proton + compact electron, so proton stability and local neutrality follow
nuclear binding	residual strong interaction	Coulomb geometry with compact electrons; He-4/deuteron ratio 13.1 against 12.7
the two masses	Yukawa couplings, free; m_p from the QCD scale	two extents of one cloud; the *ratio* geometric, the absolute scale an open problem
2. What it takes to compute it		
form of the model	a wavefunction in $3N$-dimensional configuration space with operators and probability; no continuum limit handing to the mechanics of the bulk matter atoms build	multiphase continuum mechanics in ordinary 3D — *the same physical and mathematical form as the macroscopic models*, so the hierarchy continues upward without change of character
the electron	one antisymmetrized wavefunction for all N jointly	one non-overlapping real charge density per electron, each a function of position alone
machinery	lattice QCD: Monte Carlo over a 4D path integral, $\sim 10^5$-line codes, millions of core-hours; chemistry packages with $\mathcal{O}(N^7)$ methods and fitted functionals	a Poisson solve + relaxation on a grid, linear in electron number, no basis set and no fitted parameter; the atom solver is 350 lines
what that buys	the hadron spectrum at the percent level, chemical accuracy, and quantitative reach far beyond this domain	one instrument across the entire range — nucleus, atom, molecule, and an 8×10^6-cell self-gravitating flow — at linear rather than exponential cost, running interactively on a laptop; binding and ionisation energies and nuclear binding *ratios* parameter-free; and nothing anywhere in it that can absorb a discrepancy

The last row should be read in both directions. The standard machinery is heavy because the formulations behind it are, and it buys quantitative results this construction does not reach; that is not in dispute, and simplicity is not accuracy. But neither is simplicity here mere economy. It buys *range*: no standard method carries a nucleus to a cosmic web without changing formalism at least twice, whereas here nothing changes but the constant in front of the Laplacian. It buys *size*, the cost growing linearly in electron number where a wavefunction method grows exponentially. It buys *testability*: with no basis set and no fitted exchange–correlation function there is nothing in the code that can absorb a discrepancy, so a wrong prediction surfaces as a wrong number rather than as a retuned parameter, and any reader can rerun it in an afternoon and try to break it. And it buys the paper's own argument. The claim here is that the microscopic world has the same continuum form as the macroscopic one; a solver that looks like a solver for macroscopic matter — and works — is evidence for precisely that. The simplicity is not a convenience attached to the result. It *is* part of the result.

On the cosmological constant: fixed, not absent. In the Newtonian limit of general relativity with Λ the field equation reads $\nabla^2\varphi = 4\pi G\rho - \Lambda c^2$, so Λ enters precisely as a constant subtracted from the source — the same form as (2), with $\Lambda c^2 = 4\pi G\bar{\rho}$. Historically this is no coincidence: Einstein introduced Λ in 1917 with the Neumann–Seeliger difficulty of Newtonian cosmology in view, the same difficulty that forces a prescription for $\bar{\rho}$ here. What distinguishes this model is not the absence of such a term but its *status*: in ΛCDM it is a free parameter whose near-coincidence with today's matter density is unexplained, whereas here it is fixed by vanishing total gravitational charge and tracks the matter as a^{-3}.

The Hubble diagram that follows. The coasting background is not a choice made here but a consequence of the subtraction: with no net source, $\ddot{a} = 0$ and $a \propto t$, so $H(z) = H_0(1+z)$ and

$$d_L(z) \;=\; \frac{c}{H_0}\,(1+z)\,\ln(1+z). \tag{6}$$

This is the Hubble diagram of the $R_h = ct$ class, and it is testable against supernova compilations, which is where this framework would be decided at the large scale. Published comparisons have generally favoured ΛCDM, a conclusion contested by Melia [12]. The analysis is not attempted here and the outcome is not claimed: it is recorded because (6) follows from the model whether or not the paper mentions it.

What is not claimed

The model here is **Coulomb + Newton**, and that is a real boundary rather than a form of words. Radiation is not in *this* treatment, which is electrostatic: Coulomb is the static limit of Maxwell. It is, however, already part of the framework rather than an extension awaiting invention — the time-dependent form, in which a real charge density carries a complex current and a radiating dipole, is developed in [5] and reproduces atomic spectra. Adding it is seamless in the same sense the gravitational addition is: restoring the time dependence adds terms without altering the form of the model. What follows from that for the cosmological questions below has not been worked out, and is the natural next step rather than a gap in principle.

The microwave background sits squarely in that deferred sector, and its standing here should not be misstated in either direction. It plays no role in the construction: nothing above uses it, needs it, or is calibrated against it, in contrast to ΛCDM, where it is load-bearing. But a relic radiation field is not foreign to this picture either. The sequence of Section 11 passes through a hot dense phase reached by contraction rather than by an initial singularity, and ends with matter turning neutral near 3000 K and radiation streaming free — which is a release of relic radiation by the same mechanism, differing in what produced the heat. What is not supplied is everything quantitative about it: its present temperature, the near-perfection of its blackbody spectrum, its isotropy to about one part in 10^5, and its acoustic peaks. Those are questions for the radiation sector — available in the framework, not yet turned on this problem — and they are open rather than answered.

A distinction should be drawn before any of it is attempted, because it decides how much is actually owed. That a microwave background exists, that its spectrum is Planckian to within some 5×10^{-5} of the peak (COBE/FIRAS), and that it carries anisotropies of order 10^{-5} structured into peaks in the angular power spectrum, are *measurements*. That those peaks record a last-scattering surface at $z \approx 1100$ and read off the densities of the ΛCDM parameter set is an *interpretation*, made inside a model; a different framework owes its own reading and does not inherit that one. Radiation is left over from everything that has happened, and what a relic signifies depends on what is taken to have happened — which is genuinely open. The spectrum carries the heaviest burden of all, since a Planck curve of that quality is not produced by just any process, and this paper takes no position on what produces it. The standard account reads it as thermalization in an optically thick medium; the author's own analysis of blackbody radiation [14] reaches a different conclusion, on which a Planck curve is the signature of a resonant lattice rather than of a hot disordered gas, and would therefore deny that the spectrum evidences a hot dense past at all. That question belongs to the radiation sector and is settled in neither direction here.

The same deferral applies to large-scale-structure statistics including the baryon acoustic feature, to galaxy formation, and to light-element abundances beyond helium. None is treated, and nothing above should be read as bearing on them.

The obstacles inside what *is* claimed

A limited ambition excuses what a paper does not touch, and what needs machinery it has not introduced can wait for that machinery. It does not excuse what falls *inside* Coulomb + Newton and comes out wrong. Three items are of that kind, and the first is derived from the model's own equations rather than imported from outside.

Structure cannot grow in a coasting background. The linear growth equation is $\ddot{\delta}+2H\dot{\delta} = 4\pi G\bar{\rho}\,\delta$, the source being the perturbation's own self-gravity, so the mean sets the amplitude even though only the contrast gravitates. In standard cosmology $\bar{\rho}$ and H are *locked* by Friedmann, $\bar{\rho} = 3H^2/8\pi G$, so source and friction scale alike and $\delta \propto a$: the factor 10^5 that carries 10^{-5} to order unity. Here they are decoupled — $\bar{\rho} \propto a^{-3} \propto t^{-3}$ against $H^2 \propto t^{-2}$ — so the source is subdominant at late times, leaving $\ddot{\delta} + (2/t)\dot{\delta} \approx 0$ with modes $\delta = \text{const}$ and $\delta \propto t^{-1}$: **growth freezes**. Either the mean must gravitate after all, discarding the coasting prediction, or the contrast must be *supplied* at the required amplitude rather than grown — which would make cosmic structure something the seed specifies rather than something gravity builds, and would then have to carry the observed shape of $P(k)$ as well. This is the sharpest obstacle between the present framework and a cosmology.

Which branch is taken here, and what it costs. The second branch is the one this framework points to, and the reason is worth stating because it is easy to miss. The figure 10^{-5}, which sets how much amplification is demanded, is not a measurement of any seed: it is the amplitude of the microwave-background anisotropies *read as the primordial perturbation at last scattering*, an interpretation internal to the standard model of cosmology and one this paper does not inherit (see the discussion of the background below). Without that number, the requirement changes shape. Here $\rho = \nabla^2\varphi_E$ does not describe a small ripple on top of pre-existing matter; it *is* the matter, structured at order unity from the outset. Nothing then needs to be amplified by 10^5, and gravity's role is morphology — turning order-unity contrast into filaments, sheets and voids — rather than amplitude. On the eternal reading below, where the seed is a permanent statistical property rather than an initial condition, frozen growth stops being a defect and becomes something closer to a requirement.

That is a position, not an escape, and it has a price which the preceding sections have already begun to pay. If structure is supplied rather than grown, the *shape* of the spectrum becomes the seed's responsibility, and the measurement reported in Section 9 says that responsibility is not currently discharged: the seed that collapses has the large-scale slope wrong by about two in the exponent and imposes its own correlation length as the web scale, while the seed with the right slope does not collapse. A second cost is observational. Galaxy populations *do* evolve with redshift, which is evidence of some genuine growth, and a picture in which structure is specified rather than assembled owes an account of that evolution. Both are open, and both are now questions about the seed rather than about gravitation — which is progress of a kind, in that they are well-posed and computable with the machinery already in hand.

Where that obstruction sits, and where it does not. It is worth locating precisely, because it does not lie where the new content is. Every link in the chain just given belongs to the gravitational tier: a Newtonian arena forces the Neumann–Seeliger subtraction, the subtraction removes the mean from the source, a vanishing background source gives $a \propto t$, and the consequent decoupling of $\bar{\rho}$ from H starves the growth equation. Nothing in that chain involves RealQM. Charge quantization, the neutron as $p + e$, nuclear binding from Coulomb geometry and the two masses are statements about *matter*, and are indifferent to how the background expands. The obstruction is a defect of the *arena*, not of the matter model, and it is the price of the deliberate choice to keep gravitation textbook. In general relativity no subtraction is required: the full density gravitates, Friedmann locks $\bar{\rho} = 3H^2/8\pi G$, source and friction scale alike, and $\delta \propto a$ returns. What would be surrendered is the coasting prediction (6), not any result of the preceding sections. That escape is real and is not taken here, since taking it means leaving the Newtonian arena in which the two-tier construction is posed — but it should not be mistaken for a full remedy either, because restoring growth restores neither the dark-matter amplitude nor an account of the microwave background.

Why not simply stay Newtonian on a fixed background? The arena here already *is* fixed and Euclidean: in a Newtonian setting $a \propto t$ is not expanding space but matter in free inertial expansion through it — the Milne solution, which is what a vanishing background source leaves. No expansion of space is claimed anywhere in this paper. The redshift is a **Doppler shift** from sources in relative motion, H_0 a velocity gradient rather than a property of a metric, and this is no weaker observationally: the $(1 + z)$ stretching of supernova light curves, usually cited as showing that space expands, follows equally from relativistic Doppler with the increasing light-travel time from a receding source. What differs is not the redshift but what is said to expand, and here it is the matter.

Whether the bulk motion could be dropped as well is then a fair question, and the three arenas trade against each other. With matter coasting, as here, $H = 1/t$ while $\bar{\rho} \propto t^{-3}$ decouples from it and growth freezes. With no bulk motion at all, $H = 0$ and $\ddot{\delta} = 4\pi G\bar{\rho}\,\delta$ gives exponential growth on the Jeans time, so the obstruction vanishes — but so do the Hubble law and the redshift, and a static substitute would still owe the observed $(1 + z)$ stretching, which a non-kinematic redshift does not give. With a relativistic background the mean gravitates, Friedmann relocks source to friction and $\delta \propto a$ returns, at the cost of the three claims above and the return of the dark sector. Free inertial expansion buys the Hubble law; a static box buys structure; nothing above buys both.

Two further entries, both within the model's reach. There is no extra gravitating component co-located with galaxies, so flat rotation curves and cluster mass-to-light ratios stand where Newtonian dynamics without dark matter leaves them. And $\rho = \nabla^2\varphi_E$ is a *definition* rather than a law: with no dynamics for φ_E the relation can be read backwards to yield any ρ one likes, so the seed is an input here — as the primordial spectrum is an input in the standard account — but without the processing that would connect an input spectrum to an observed one, which is what the test of Section 9 showed to be missing.
None of this is addressed here. That is why the paper claims an extension to gravitation and not a cosmology.

A feed, and a zero sum

What would buy both: a continuous feed. Neither of the arenas above yields the Hubble law and structure formation together. One modification would, and it is worth recording because it is cheap to state and expensive only where it should be. Suppose φ_E is not given once but *fed* — sourced continuously, so that curvature, and with it charge and matter, is replenished as fast as the expansion dilutes it. Holding $\bar{\rho}$ fixed requires the rate $3H\bar{\rho} = 3\bar{\rho}/t$, which is Hoyle's. The growth equation then reads $\ddot{\delta} + (2/t)\dot{\delta} = 4\pi G\bar{\rho}\,\delta$ with a *constant* source against a friction that decays, so at late times $\delta \propto \exp(t/\tau)$ with $\tau = (4\pi G\bar{\rho})^{-1/2}$: structure grows exponentially on the Jeans time while the expansion still coasts.

What makes this available here, and not in the standard account, is the very decoupling that caused the trouble. Friedmann locks $\bar{\rho}$ to H, so a changed density is a changed expansion; here the subtraction has already severed that link and only the contrast gravitates, so $\bar{\rho}$ can be held up without touching $a(t)$. The feature that starves growth is what makes the remedy possible. Two consequences come with it. With $\bar{\rho}$ constant the subtracted term $\Lambda c^2 = 4\pi G\bar{\rho}$ is a genuine constant rather than something tracking a^{-3}; and creation occurs in $\pm$ pairs as the curvature of one potential, so charge balance is automatic rather than an added condition — which was not the position of Hoyle, Bondi and Gold, who needed matter to appear as matter.

It is not adopted here, and the reasons deserve stating as plainly as the mechanism. Nothing motivates the feed except the work it does: φ_E acquires a source term the model does not otherwise need, at a rate fixed by the requirement that it succeed. It moves the eternal reading below towards a steady state and inherits that model's difficulties, the observed evolution of source populations with redshift above all. It is recorded as the one known way to have the Hubble law and structure formation together in this arena — and as something to compute rather than an answer.

What the feed would cost in energy: nothing, if the sum is zero. The obvious objection to a feed is energetic, and it has a definite answer. The rate required is small in the only units one can picture: with $H_0 = 2.2 \times 10^{-18}\,\mathrm{s}^{-1}$ and a baryon density $\bar{\rho} \approx 4 \times 10^{-28}\,\mathrm{kg\,m^{-3}}$, $3H\bar{\rho} \approx 3 \times 10^{-45}\,\mathrm{kg\,m^{-3}s^{-1}}$, about *one hydrogen atom per cubic metre per* 2×10^{10} *years.* Small per unit volume; by construction it is the entire matter content per Hubble time, so any test of it is cosmological and not local.

Where the energy comes from admits a sharp answer, and it is not from collapse. Gravitational collapse converts at efficiency GM/Rc^2 — about 3×10^{-6} for a galaxy, $\sim$0.2 for a neutron star — which is less than unity always, so collapsing mass M never yields the mc^2 needed to create mass m unless $M > m$. Gravity recycles binding energy from matter that exists (Section 11); it cannot be a net source of matter. The zero-energy route is different and does work: matter is created *together with* its own negative potential energy, a new pair carrying $+mc^2$ of rest energy and $U \approx -GMm/R$ against everything else, so that if $GM/Rc^2 \approx 1$ the pair costs nothing and no collapse need fund it.

The idea is old and is not claimed here: Tryon [13] proposed the universe as a zero-net-energy fluctuation, and inflationary cosmology invokes the same ledger as its "free lunch". It is worth being exact about its standing. In general relativity the statement is sharp only in a particular setting — for a spatially closed universe the Hamiltonian constraint gives exactly zero — while the Newtonian form used here is heuristic. And it is no part of the Standard Model, which is a field theory on a fixed background containing no gravitation at all, so nothing in it can balance a rest energy against a potential one.

Whether the balance can hold here at all is a quantitative question, and the answer is presently negative. Two things have to be checked, and both fail in the same direction.

Permanence. GM/Rc^2 is fixed in time only if $M \propto R$, hence only if $\bar{\rho} \propto t^{-2}$. That is what Melia's $R_h = ct$ delivers, but it delivers it *through* Friedmann, which ties $\bar{\rho}$ to H^2 — the relation this paper does not have. Here matter dilutes as $\bar{\rho} \propto a^{-3} \propto t^{-3}$ while $R = ct$, so the enclosed mass is constant and $GM/Rc^2 \propto 1/t$ *decreases*: the books balance at one epoch at most, not permanently. A feed at the rate $H\bar{\rho}$ — a third of the rate that holds $\bar{\rho}$ fixed — would give $\bar{\rho} \propto t^{-2}$, hence $M \propto R$, hence a constant ratio, and would still restore growth, now as a power law rather than exponentially. That is the version of the feed the ledger asks for.

Magnitude. At critical density $GM/Rc^2 = \frac{1}{2}$, which is the coincidence usually quoted. But this model has no dark matter, so its density is the baryon density, $\Omega_b \approx 0.05$, giving $GM/Rc^2 \approx 0.025$ — short of unity by a factor of twenty to forty. The positive side of the ledger is the rest energy $(m_p + m_e)c^2 = 938.8$ MeV per pair, carried by structure at 10^{-15}–10^{-10} m, where charge fails to cancel; the negative side is gravitational and global, and in a universe of baryons alone it is too weak by that factor to cancel it. The zero-energy condition therefore calls for roughly twenty times more gravitating mass than the model contains — the same factor demanded by rotation curves and by structure formation, arrived at independently.

Two limits should be stated rather than left for a reader to find. First, the ledger uses the *mean* density through M, which the dynamics of (2) subtract; these are two regularizations of one Neumann–Seeliger divergence and not a contradiction — the subtraction concerns *force*, which a uniform medium does not exert by symmetry, while the energy accounting concerns a horizon cutoff, the mass within *ct* — but the two must be kept distinct in the same breath. Second, the cancellation is order-unity and not exact: the self-energy may be written $-GM^2/R$ or $-3GM^2/5R$ and the cutoff taken at the Hubble radius or the particle horizon, which moves the required density by factors of a few about the critical value. Exactness would require a definite prescription for both, and the framework does not yet supply one.

An eternal universe as the less brutal alternative. If the seed is not an initial condition but a **permanent statistical property** of φ_E, there is no first moment to account for: the universe is *statistically stationary* — ensemble properties fixed, contents forever different — rather than the steady state of Hoyle, Bondi and Gold, which needs continuous creation to hold the density constant — though the feed considered above would move it in exactly that direction. This changes the standing of the growth obstruction, since with infinite time available unbounded growth would be inadmissible, so frozen growth becomes closer to a requirement than a defect. But eternity is not free: Tolman's entropy accumulation needs a recycling mechanism and none is offered, Olbers' paradox is eased but not dismissed, the observed evolution of source populations with redshift must be accommodated, and a coasting expansion has a *finite* age $1/H_0 \approx 14.5\,\text{Gyr}$, so eternity would require a bounce or cycles, not supplied here.

13 Conclusion: two tiers, one two-valued Poisson form

The construction is complete — a perturbation of φ_E quantized into charges, Coulomb building nuclei, atoms and molecules from them, Newtonian gravitation acting on the energy of what is built — and an observation can now be recorded that was used as a premise nowhere above. Each tier was set up on its own terms; that is what makes the observation worth recording.

The two tiers, arrived at independently, turn out to have *one* structure and to differ in a single sign. Each force is the gradient of a potential obeying Poisson's equation with a two-valued source,

$$\text{electric:} \quad \rho_c = -\varepsilon_0\, \nabla^2 \varphi_E, \qquad\qquad \text{gravitational:} \quad \rho_m = \frac{1}{4\pi G}\, \nabla^2 \varphi_g, \tag{7}$$

the force being $-(\text{charge})\,\nabla\varphi$ in each case. The one structural difference is the relative sign, and with it the force law:

	like sources	unlike sources
electric (charge)	**repel**	attract
gravitational (mass)	**attract**	repel

There are thus two independent binary signs — the source $\pm\nabla^2\varphi$ and the force $\mathbf{F} = \pm\nabla\varphi$ — whose four combinations are exactly the four "charges" the two forces couple to: the repulsive-for-like sign is electromagnetism, its $\pm$ source the $\pm$charge of the small scale; the attractive-for-like sign is gravitation, its $\pm$ source the $\pm$ contrast of the large. One operator and two signs yield $\{\pm\text{charge}, \pm\text{mass}\}$.

That single sign difference generates the two tiers. Because like charges repel, matter neutralises into $\pm$ pairs and electromagnetism *screens*, staying local and small-scale. Because like masses attract, mass *accumulates* into large-scale structure, and because unlike sources repel, overdense and underdense regions separate into web and voids. The scale separation is thus a *consequence* of the force-sign, not a separate assumption. A second asymmetry — the *direction* in which the Poisson relation is used — fixes the scale of each force. For electromagnetism the potential is primary and charge is obtained by *differentiating*: the Laplacian is local and magnifies short wavelengths ($\rho_k \propto k^2\varphi_k$), so a small-scale perturbation is inflated into sharp charges. For gravitation the source is primary and the potential is obtained by *integrating*: the inverse Laplacian is global and smooths ($\varphi_k \propto \rho_k/k^2$), building a broad potential. The same relation run forward makes small-scale electromagnetism and run backward large-scale gravitation — and that is why charge is small and gravity large.

One practical consequence of the shared form deserves recording alongside the formal one, since it is what made the chain computable end to end rather than merely statable: sharing an operator, the two tiers also share a solver. The same Poisson-plus-relaxation primitive on the same kind of grid produces a free-boundary atom and an 8×10^6-cell self-gravitating flow, with no change of formalism and no fitted parameters at either end. Whether the framework is right is decided by the physics; that it can be *followed*, scale by scale and without a change of instrument, is decided by this.

This is a *structural parallel*, observed after the fact, and we resist promoting it into a premise. It does not say gravitation is derived from electromagnetism, or that one seed generates both sectors: the gravitational source here is the matter's own energy density and owes nothing to φ_E beyond the matter it made. Nor does the shared form make the two constants one — whether α and α_G reduce is the open question below, and the parallel is a reason to ask it, not an answer. What can fairly be said is that two forces spanning forty orders of magnitude in strength, introduced for entirely separate reasons — one to make matter, one to gather it — were found to be the same equation read with two signs.

14 Open problems

1. **The proton–electron size difference**, in three parts of unequal status. *(i) The ratio* is read here as geometric: a caged, zero-localisation electron sits at R_p and a free one at a_0, so $a_0/R_p \approx 1/\alpha^2$, needing no relativity. This the model addresses. *(ii) The absolute scale* of R_p — the same datum as the mass ratio and the nuclear energy scale: one open number, not derived here, nor in the Standard Model. *(iii) The sign asymmetry*: why + is the compact sign at all. Pure Coulomb with the Laplacian is charge-conjugation symmetric, so the asymmetry cannot arise from these ingredients; a natural candidate is a *non-Gaussian (skewed)* seed, whose super- and sub-level sets are not mirror images — one sign forming sharp compact pits, the other broad swells — a geometric route from the *statistics* of the seed rather than a new force.

2. **Could the two potentials be one field?** Split a single potential into scales, $\varphi = \varphi_{\text{short}} + \varphi_{\text{long}}$, reading charge as the short-wavelength Laplacian and mass contrast as the long-wavelength one. The objection that a neutral hydrogen cloud has vanishing charge but non-vanishing mass does not apply, the two roles never being evaluated at the same scale. Two conditions would have to be met: a filter scale fixed by something physical, presumably where matter condenses; and the abundance of small-scale structures tracking $\nabla^2\varphi_{\text{long}}$, which

for a Gaussian seed fails identically and requires local-type non-Gaussianity — the same skewness wanted in item 1(iii), so one assumption might serve both. This would also be what it takes for the seed to supply the large-scale inhomogeneity that is currently an input. Whether a skewed seed's small-scale power correlates with its own long-wavelength Laplacian, and with which sign, is a well-posed calculation, not attempted here.

3. **The fluctuation spectrum and a law of motion.** $\rho_E = -\varepsilon_0 \nabla^2 \varphi_E$ is a definition, not a dynamics: φ_E has no equation of motion, and its spectrum is an input at the small scale just as the matter spectrum is an input at the large one (Section 10). These are two separate gaps and the second is the load-bearing one: the shape as well as the amplitude of the large-scale contrast is supplied, so the model does not predict $P(k)$. Closing it would need the seed's long wavelengths to modulate its short-wavelength power, as in item 2 above; short of that, supplying φ_E with a law of motion would close only the first.
4. **The mass ratio $m_p/m_e = 1836$, and why it is not the nuclear scale.** The mass ratio implies a nuclear size $a_0/(m_p/m_e) \approx 29$ fm, whereas the actual scale is ~2–3 fm; equivalently the size ratio $a_0/R_p \approx 2.5 \times 10^4 \approx 1/\alpha^2$ exceeds 1836 by a factor ~14. The nucleus is that much more compact than the naive size–mass law allows, because the caged electron is flat, held collectively by several protons. Deriving that factor would make m_p/m_e fix the nuclear scale, hence α, and collapse two numbers into one; asserting it incidental would be numerology.
5. **Antimatter.** The $\pm$ signs of the seed are the proton and the electron, both matter, so the matter–antimatter asymmetry of the standard account does not arise (Section 5). The debt this incurs is the mirror question: a positive cloud of electron extent is a positron, nothing in the construction forbids it, and why such clouds are not abundant is not explained here. Pair production, and the fate of the positrons it makes, lie in the time-dependent (Maxwell) sector rather than the electrostatic one treated here.
6. **The neutrino.** β-decay's continuous spectrum — the 1930 objection that sank the proton–electron neutron [6]. The "electron expands" step needs a home for that energy and spin; an attempt to have a field carry it came out too weak, so the neutrino is neither derived nor eliminated here.
7. **Precision and scope.** Abundances beyond helium, fluctuation-to-structure statistics, and the extreme regimes (strong-field gravity, the highest energies) lie beyond the Coulomb+Newton scope kept here.

The count of constants. Only dimensionless numbers are physical. Stripped to essentials the model carries **three**:

$\alpha = 1/137$	Coulomb strength	sets the size hierarchy, $a_0/R_p \approx \alpha^{-2}$
$m_p/m_e = 1836$	the two charge species	sets the mass hierarchy
$\alpha_G = Gm_p^2/\hbar c \approx 5.9 \times 10^{-39}$	gravitational strength	sets where Newton takes over

The first two are both small-scale: atoms, molecules, chemistry and nuclear binding ratios follow from them parameter-free, gravity playing no part. The third governs the transition between the tiers, and does so at a definite scale: a body can be held up electromagnetically only up to about $\alpha_G^{-3/2} \approx 2 \times 10^{57}$ protons $\approx 1.8\, M_\odot$, the stellar mass scale, beyond which gravity wins. The smallness of α_G is therefore not an embarrassment but the very thing that creates the two-tier structure: gravity is so weak that matter can be chemical over an enormous range before Newton asserts itself. Against this, the Standard Model carries some nineteen free parameters and standard cosmology adds several more. Three dimensionless numbers is a striking reduction — but only *over the domain covered here*, and that qualification is doing real work, so it is spelt out rather than left to the reader.

Claimed here, from α and m_p/m_e	**Not attempted; the Standard Model's**
atomic and molecular structure, chemistry	quark substructure and QCD
nuclear binding by Coulomb geometry, no strong force	the particle spectrum, generations, mixings
charge quantization ± 1 as a consequence	precision QED (e.g. $g-2$ to ten digits)
local neutrality; no baryogenesis required	the weak sector (freeze-out is borrowed, not derived)
the neutron as $p+e$, hence proton stability	anything relativistic or at collider energies

The right-hand column is not a list of things this model gets wrong; it is a list of things it does not address at all, and it is longer and heavier than the left. A parameter count is only meaningful together with the domain it covers, and on that measure the Standard Model buys its nineteen parameters a great deal more than three buy here. What can be said is narrower and still worth saying: *within* the domain of atoms, molecules and nuclear binding *ratios*, this construction is parameter-free where the standard account is not, and it derives two things — charge quantization and local neutrality — that the standard account must supply.

To that, three honest caveats. α_G is not derived (it is the hierarchy problem, and no one has derived it); m_p/m_e and $1/\alpha^2$ differ by the unexplained factor $\sim$14 of item 4; and whether α and G themselves reduce to one is the deeper question this deliberately parallel construction does not settle.

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